The empirical recurrence for OEIS A255778, recovered from its tail

Adrian Perez Fontelles
Independent researcher

8 September 2026

Abstract

OEIS A255778 records "Empirical recurrence of order 78" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 132$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 6$, so the recurrence is proved for every $n \geq 84$. Its last coefficient is $c_{78} = -18336 \neq 0$, so no further tail satisfies anything shorter; any order-78 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A255778 is "Number of (n+2)X(3+2) 0..1 arrays with no 3x3 subblock diagonal sum 1 and no antidiagonal sum 1 and no row sum 0 and no column sum 0.". It has offset 1 and begins

$$1512, 8672, 58729, 327340, 1968584, 11945779, 71461994, 428975207, \ \ldots$$

The entry states:

> Empirical recurrence of order 78 (see link above)

As of the "Last modified" line on the live entry (Jun 02 2025 11:41:38, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 132$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 132$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 78: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 6$ is the first whose minimal order is exactly the stated 78.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 264$ exact terms from $n = 6$ on, it returns order 78 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{78} c_i\, a(n-i),$$

c_1	4	c_{21}	34497535	c_{41}	4041531822	c_{61}	−975600266
c_2	12	c_{22}	48559283	c_{42}	−1837949905	c_{62}	−205381174
c_3	8	c_{23}	3161880	c_{43}	2005922569	c_{63}	−218641318
c_4	61	c_{24}	−108896739	c_{44}	−970749583	c_{64}	465915072
c_5	−414	c_{25}	−138466994	c_{45}	570919190	c_{65}	91607298
c_6	−453	c_{26}	−143602164	c_{46}	−1436258296	c_{66}	−160799434
c_7	−4602	c_{27}	97630556	c_{47}	−2754926011	c_{67}	38740168
c_8	−7654	c_{28}	371084053	c_{48}	983355915	c_{68}	−31485230
c_9	33012	c_{29}	337828163	c_{49}	−2865469314	c_{69}	57260478
c_{10}	61110	c_{30}	−309381417	c_{50}	4113461854	c_{70}	−23691900
c_{11}	67240	c_{31}	−593568183	c_{51}	−304509836	c_{71}	−132156
c_{12}	−39778	c_{32}	−1172531253	c_{52}	−1358263559	c_{72}	6661192
c_{13}	−149863	c_{33}	971314515	c_{53}	2632191900	c_{73}	−1768368
c_{14}	−234361	c_{34}	458244748	c_{54}	−320225605	c_{74}	922368
c_{15}	−1293983	c_{35}	84625069	c_{55}	1067153629	c_{75}	−827376
c_{16}	−960594	c_{36}	2997725665	c_{56}	−617286215	c_{76}	324656
c_{17}	−1900233	c_{37}	−2214227134	c_{57}	−780085524	c_{77}	42720
c_{18}	−503735	c_{38}	−964972099	c_{58}	−576414218	c_{78}	−18336
c_{19}	3899697	c_{39}	−1693255474	c_{59}	506399596		
c_{20}	6899395	c_{40}	2632646	c_{60}	783805673		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 84$, and it is the recurrence OEIS A255778 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 6$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{78} - c_1 x^{78-1} - \cdots - c_{78}$. Its constant term is $-c_{78} = 18336$, which is not zero, so $q_{\min}(0) \neq 0$ and

$q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 78 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 78, so $\deg q = 78$, and it is monic. It holds from some index t on. From $\max(t, 6)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 78, so $q = q_{\min}$. □

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 18 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 78, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -18336.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-78 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A255778.

[2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)

[5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.